

CS 217: Practice questions on Generative Classifiers

- Write the SVM **dual** objective for the separable case where we do not have any slack variables, that is, for this objective:

$$\begin{aligned} \min_{\mathbf{w}, w_0} \quad & \frac{1}{2} \|\mathbf{w}\|^2 \\ \text{s.t.} \quad & y_i(\mathbf{w} \cdot \mathbf{x}_i + w_0) \geq 1 \quad \forall i : 1 \dots N \end{aligned} \tag{1}$$

..3

$$\max_{0 \leq \alpha_i, \sum_{i=1}^N \alpha_i y_i = 0} -\frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j y_i y_j \mathbf{x}_i \cdot \mathbf{x}_j + \sum \alpha_i$$

- Consider a trivially separable dataset on $d = 3$ dimensions consisting of five instances and their labels as: $D = \{([1, 2, 1], 1), ([1, 1, 1], 1), ([0, 0, 0], -1), ([-3, -1, -2], -1), ([-2, -1, -1], -1)\}$.

- If you use the linear SVM max-margin formulation for the above separable case (i.e., without the slack variables ξ_i), what are the support vectors for this dataset? ..1

The points $[1, 1, 1]$ and $[0, 0, 0]$

- For the above case, what is the optimal value of parameters $\mathbf{w} = [w_1, w_2, w_3]$ and b ? ..3 We want to solve for $\min_{\mathbf{w}} \|\mathbf{w}\|$ s.t. $y_i(\mathbf{w} \cdot \mathbf{x}_i + b) \geq 1$. It is easy to see that this is minimized by choosing $\mathbf{w} = [2/3, 2/3, 2/3], b = -1$.

- If you now change the objective to minimize the L1 norm ($|w_1| + |w_2| + |w_3|$) instead of L2 norm ($w_1^2 + w_2^2 + w_3^2$), describe all possible optimal solutions for w_1, w_2, w_3 and b ..2 Any positive w_1, w_2, w_3 s.t. $w_1 + w_2 + w_3 = 2$ suffices.

- Draw a compact decision tree on the above training data. ..2 Any single attribute can perfectly separate the data. So, a single node tree on any of $x_1 \leq 0, x_2 \leq 0, x_3 \leq 0$ will work.

- Suppose on this dataset you trained various classifiers optimally on D and now we add one more instance $(\mathbf{x}^6, y^6) = ([3, 3, 3], 1)$. For which of the classifiers below will the parameters $\mathbf{w}, b$ change after retraining with these six instances. Give a brief justification for each.

- SVM for the separable case above. ..1 No change since this point is not SV. No change since loss continues to be zero.

- Naive Bayes classifier ..1 Will change

- If you find that SVM with a RBF kernel is over-fitting your data. State with a brief reason which of the following actions you will take:

- Increase or decrease C ? ..1 Decrease C

- Increase or decrease σ of the RBF kernel $e^{-(\frac{\mathbf{x}^i - \mathbf{x}^j}{\sigma})^2}$? ..2

If distances die down quickly changes of over-fitting is higher. Thus, σ has to be decreased for a region to be influenced by multiple points.

4. Consider a trivially separable dataset on $d = 2$ dimensions consisting of four instances and their labels as: $D = \{([1,1],1),([-1,-1],-1),([1,2],1), ([-3, -1], -1)\}$. Suppose we train a SVM on this dataset with a linear kernel and large C .

- (a) Show geometrically the training points in 2D space along with two separators — one which maximizes the separation between points of the two classes, and a second which does not. ..2
- (b) What is the optimal value of parameters $\mathbf{w} = [w_1, w_2]$ and b assuming C is large? ..2
Since the data is separable and C is large, we can set $\xi_i = 0$. Next, we want to solve for $\min_{\mathbf{w}} ||\mathbf{w}||$ s.t. $y_i(\mathbf{w}\mathbf{x}^i + b) \geq 1$. It is easy to see that this is minimized by choosing $\mathbf{w} = [0.5, 0.5], b = 0$.
- (c) What are the support vectors for this dataset? ..1 The points $[1,1]$ and $[-1, -1]$
- (d) What is the value of α_i for these four points? [Hint: Do you try to solve exhaustively. Use the answer to the above question to simplify, but show the steps you followed instead of just the final answer.] ..2 Only two α_i s are non-zero, ($\alpha_3 = \alpha_4 = 0$) and they are of opposite labels. This implies that $\alpha_1 = \alpha_2$. We can solve for this α in closed form.
- (e) If you now change the training objective to minimize L1 norm of $\mathbf{w}$ instead of L2 norm of $\mathbf{w}$. Note, for $d = 2$, L1 norm will be $|w_1| + |w_2|$ and L2 norm will be $w_1^2 + w_2^2$, describe all possible optimal solutions for w_1, w_2 and b ..2 Any positive w_1, w_2 s.t. $w_1 + w_2 = 1$ suffices.
- (f) If you now change the Hinge loss by a perceptron loss defined as follows:

$$\min_{\mathbf{w}, b} C \sum_i \max(0, -y_i(\mathbf{w}\mathbf{x}^i + b)) + ||\mathbf{w}||^2$$

what is the optimal value of the parameters $\mathbf{w} = (w_1, w_2), b$ (Assume $C > 0$)? ..2
 $w_1 = w_2$ a small positive value.

- (g) If you replace the perceptron loss above by the logistic loss, does the solution that you found above continue to be optimal? State a brief reason for your answer? [Assume $C = 1$] ..2 No, for the logistic loss we want $\log(1 + e^{-y_i(\mathbf{w}\mathbf{x}^i + b)})$ to be close to zero. For that we want $\mathbf{w}\mathbf{x}^i + b$ to be large in magnitude and of same sign as y_i . For example if we pick $\mathbf{w} = [1, 1], b = 0$ we get a lower objective value than the one with perceptron which would be close to $4 \log 2$.

5. Exercise 17.1 in Book: “Probabilistic Machine Learning” by Kevin Murphy.